EXNO 5	Setting, Configuring and Managing Three Password Policies in the Computer (**BIOS, Administrator and Standard User**)
Date:	

Aim

To set, configure, and manage password policies at three levels in a computer system: **BIOS password, Administrator password, and Standard User password** for improving system security.

Requirements

- Computer / Laptop
- Operating System: **Microsoft Windows**
- System BIOS / UEFI setup
- Administrator access

Theory

A password policy is a set of rules used to enhance computer security by encouraging users to create strong passwords and store them securely. Password protection can be applied at different levels of a computer system.

1. **BIOS Password** – Protects the system before the operating system loads. It prevents unauthorized users from changing system settings or booting the computer.
2. **Administrator Password** – Provides full control over the operating system, including installing software, changing system settings, and managing users.
3. **Standard User Password** – Used for normal users with limited permissions, preventing them from making system-level changes.

Implementing these three levels of password protection improves overall system security.

Procedure –

1. Setting BIOS Password

- **Restart the Computer**
  - Click the **Start Menu** in **Microsoft Windows**.
  - Select **Power → Restart**.
  - The computer will shut down and start again.
- **Press the BIOS Setup Key During Startup**

    - While the computer is starting, press the BIOS setup key repeatedly.
    - Common keys used to enter BIOS are **F2, F10, Delete (Del), or Esc**.
    - The correct key is usually displayed on the screen as **"Press F2 to enter Setup"** or **"Press Del for BIOS Setup."**
- **Enter the BIOS Setup Utility**
    - After pressing the correct key, the system will open the **BIOS Setup Utility**.
    - BIOS is the firmware that controls hardware settings before the operating system loads.
    - You will see several menus such as **Main, Advanced, Security, Boot, and Exit**.
- **Navigate to the Security or Password Section**
    - Use the **arrow keys** on the keyboard to move between menus.
    - Select the **Security** tab or **Password Settings** option.
    - This section manages password protection for the system.
- **Select "Set BIOS Password" or "Admin Password"**
    - Choose **Set admin Password** or **Set BIOS Password**.
    - The **admin Password** allows the administrator to control BIOS settings.
    - Some systems also provide **User Password** for limited access.
- **Enter and Confirm the Password**
    - Type a **strong password** using letters, numbers, and symbols if allowed.
    - Press **Enter** after typing the password.
    - Re-enter the same password again to confirm it.
- **Save the Password Settings**
    - After setting the password, press **F10** or select **Save & Exit** from the Exit menu.
    - A confirmation message such as **"Save configuration changes and exit?"** will appear.
    - Select **Yes** and press **Enter**.
- **System Restart and Password Protection**
    - The computer will restart automatically.
    - From the next startup, the system will ask for the **BIOS password** before allowing access to BIOS settings or boot options.

2.Setting Administrator Password in Windows

- Log in to Windows.
- Open **Control Panel**.
- Select **User Accounts**.
- Click **Manage another account**.
- Select the **Administrator account**.
- Click **Create a password**.
- Enter a strong password and confirm it.
- Click **Create Password** to save.

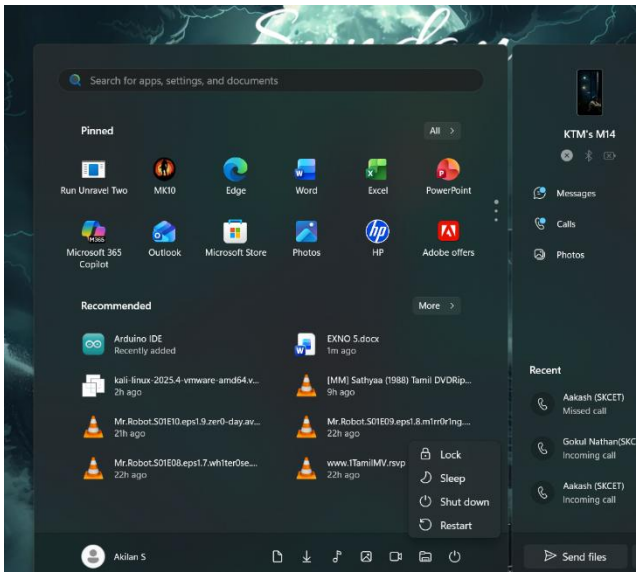
3. Creating Standard User Account and Password

- Open **Control Panel**.
- Select **User Accounts**.
- Click **Manage another account**.

- Select **Add a new user in PC settings**.
- Create a new account with **Standard User** access.
- Enter the username and password.
- Save the settings.

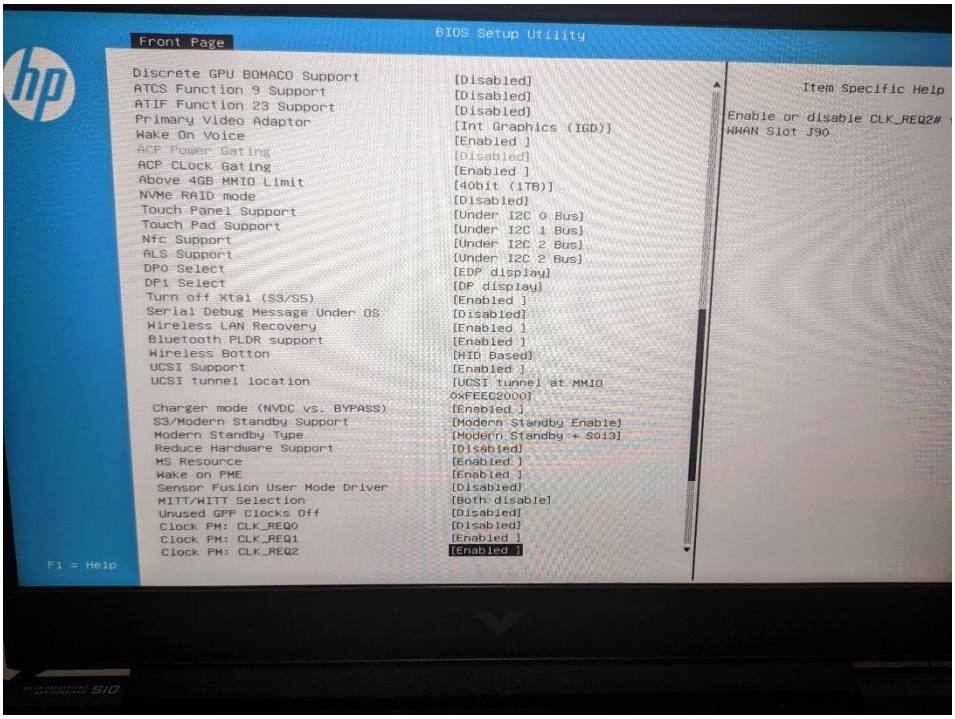
IMPLEMENTATION OF BIOS

1.Restart the Computer:



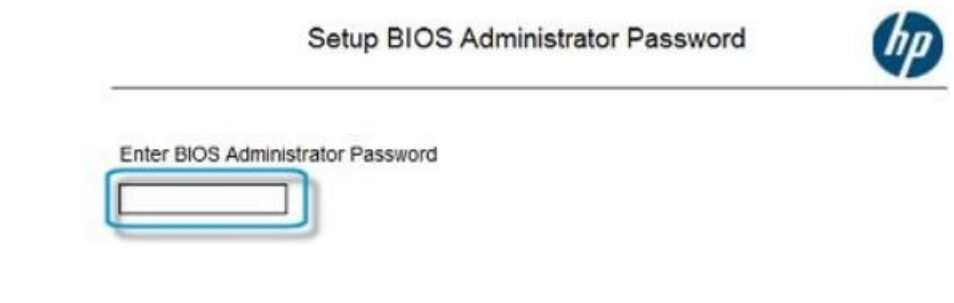
2.Press the BIOS Setup Key During Startup

3.Enter the BIOS Setup Utility

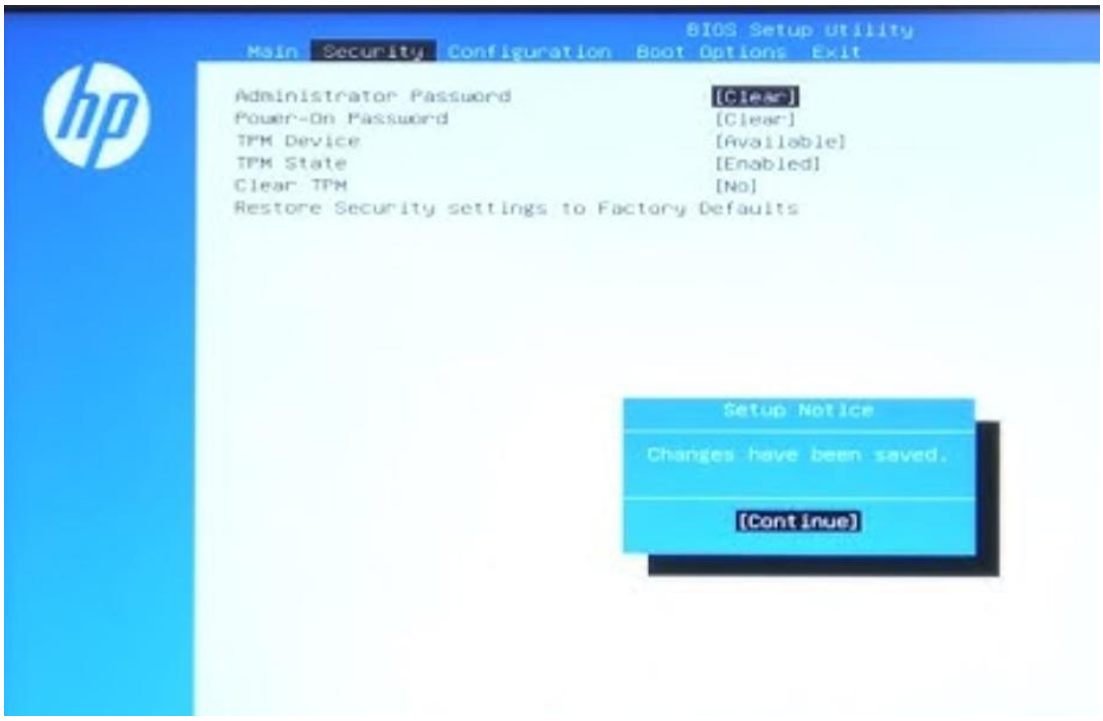


4.Navigate to the Security or Password Section

4. **Select “Set BIOS Password” or “Admin Password”**



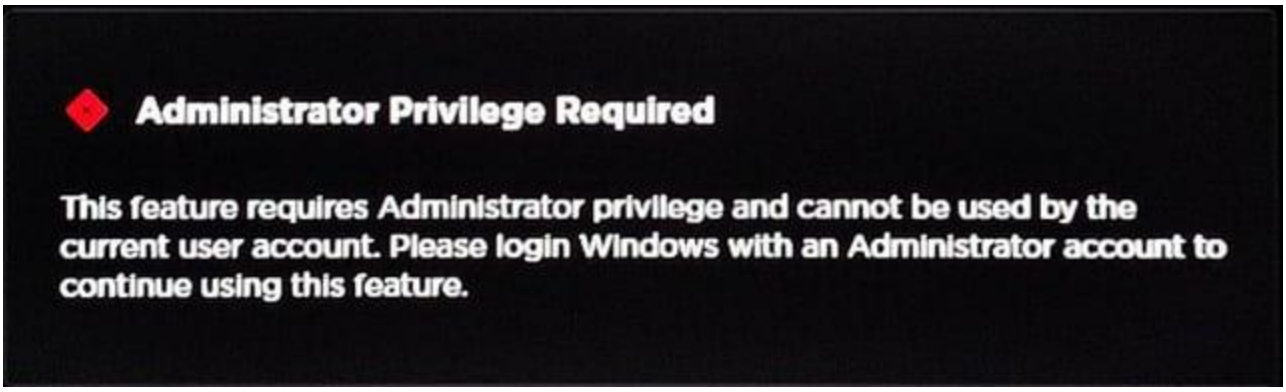
After setting BIOS password :



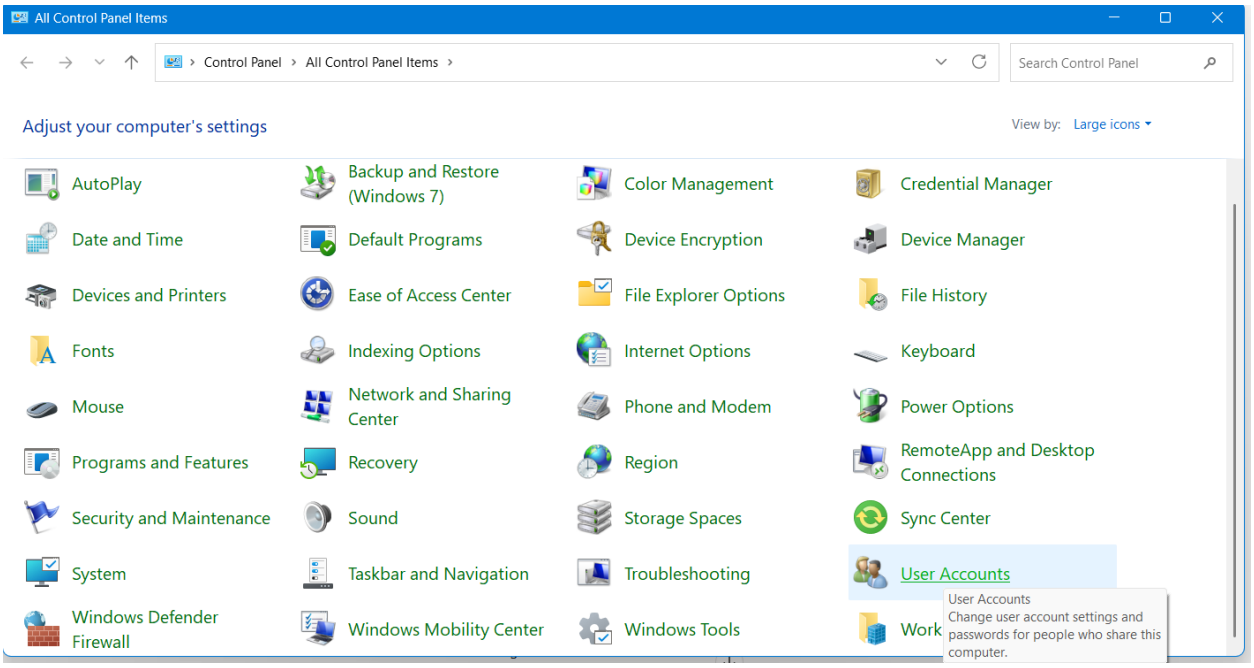
Difference between User control and Admin Control:

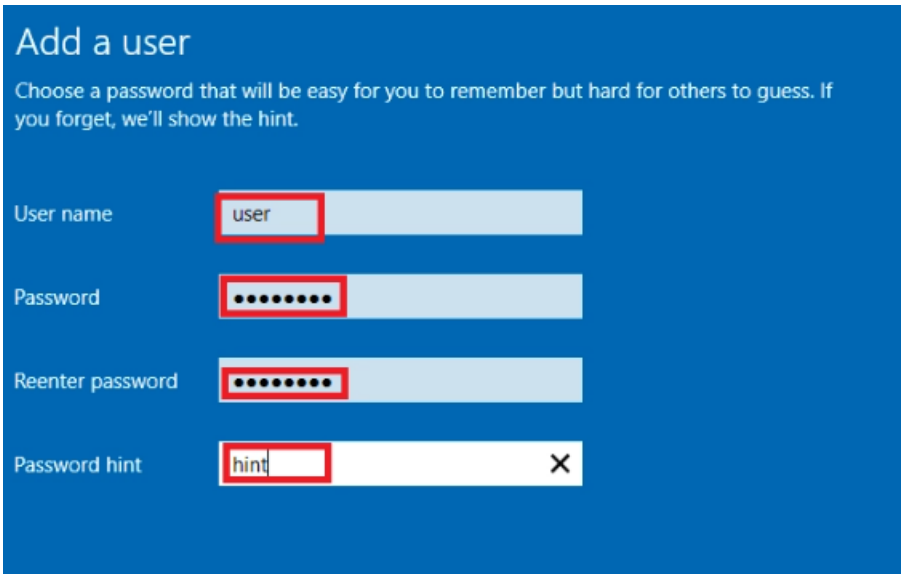
Feature	User Password	Administrator Password
Purpose	Used to start the computer	Used to control and manage BIOS settings
BIOS Access	Cannot change BIOS settings	Can access and modify BIOS settings
Permission Level	Limited access	Full access
Security Control	Only allows system boot	Controls entire BIOS configuration
Who Uses It	Normal user	System administrator

User Control Preview:

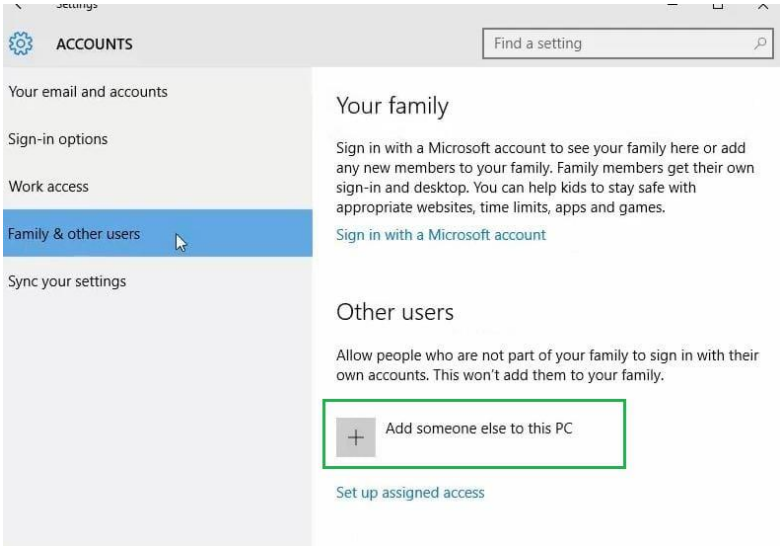


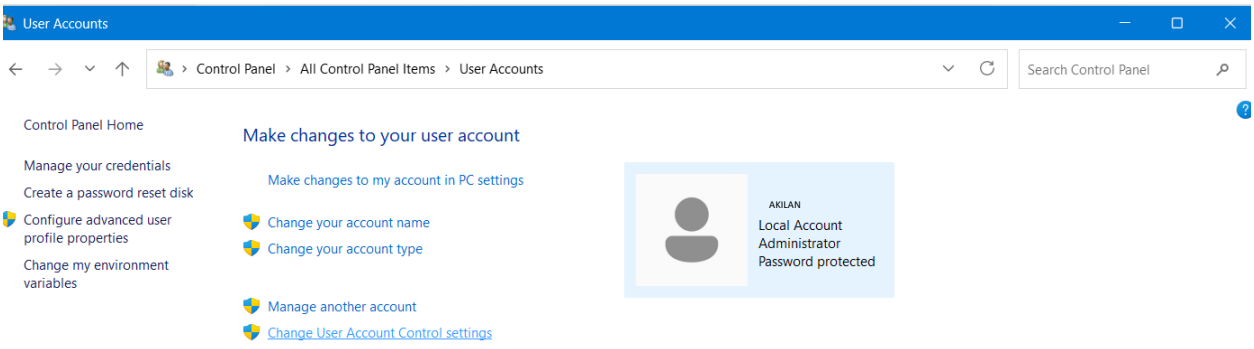
Admin Control Preview:

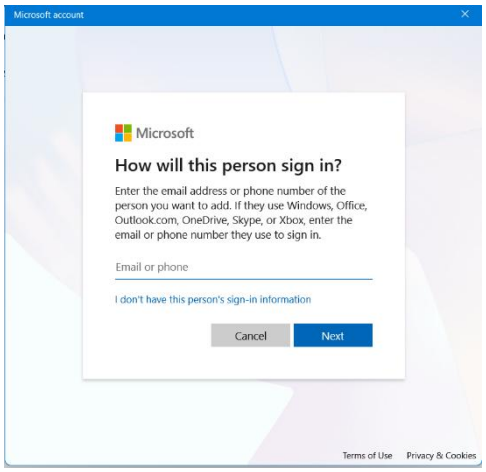


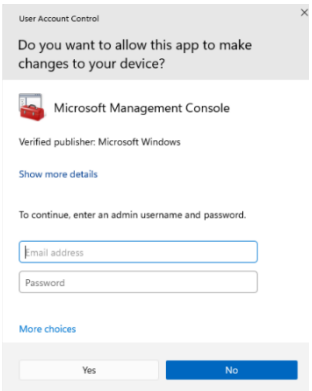


2. Setting Administrator Password in Windows And Creating Standard User Account and Password

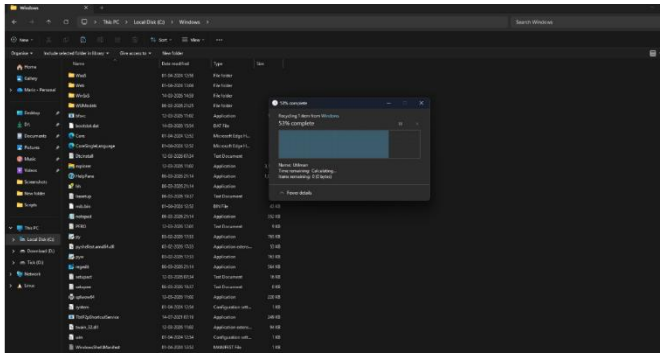








Try to delete in user



Try to delete in admin

Result

The **BIOS password, Administrator password, and Standard User password** were successfully set and configured to improve system security.